

Course: SER216

Student Name: Elyor Ataboyev

Project: Digital Parking Permit System

GitHub Repository: https://github.com/Elyor200/SER216_Final_Project_ElyorAtaboyev.git

Task 1: Requirements

Functional Requirements:

1. The system shall allow students and staff to register vehicle details (make, model, license plate number) linked to their university ID.
2. The system shall enable users to submit a parking permit request for a specific duration and permit type.
3. The system shall display real-time availability of parking zones based on the user's permit eligibility.
4. The system shall allow security staff to scan/verify permits via mobile device and log parking violations for unauthorized vehicles.

Non-Functional Requirements:

1. **Performance:** The permit verification process at security checkpoints must complete within 2 seconds under normal system load.
2. **Security:** All personal user data and vehicle registration information must be encrypted at rest and in transit using AES-256 encryption standards.

Task 2: Workflow Diagram



Workflow Description:

- 1. **Vehicle Registration:** User inputs vehicle details into the system.
- 2. **Permit Request:** User submits a request for a parking permit.
- 3. **Zone Check:** System verifies eligibility and displays available parking zones.
- 4. **Permit Approval:** System or admin approves the request and issues the digital permit.
- 5. **Parking Verification:** Security staff scans the permit to validate access or log violations.

Task 3: Test Case Table

Test ID	Feature	Input / Data	Expected Result	Test Type
---------	---------	--------------	-----------------	-----------

TC-01	Integrated Permit & Zone Check	Valid user requests permit for Zone A	System assigns permit and decreases Zone A capacity by 1	System Test
TC-02	Security Verification	Security scans an invalid/unregistered license plate	System flags violation and updates violation record	System Test
TC-03	User Registration	Staff member registers vehicle with valid University ID	Account created and staff-specific permit options displayed	Acceptance Test
TC-04	Expiration Notification	Permit expiration date is within 3 days	User receives automated email notification to renew permit	Acceptance Test
TC-05	Vehicle Registration Module	Submit new vehicle details after recent DB schema update	Vehicle saved correctly without data loss from previous modules	Regression Test
TC-06	Permit Verification Logic	Scan a known valid, unexpired permit	System displays "Valid" and does NOT flag as expired	Regression Test

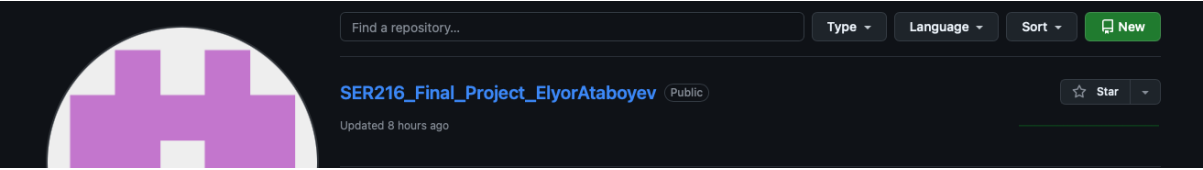
Task 4: Defect Analysis

Defect: "A valid parking permit is shown as expired during security verification."

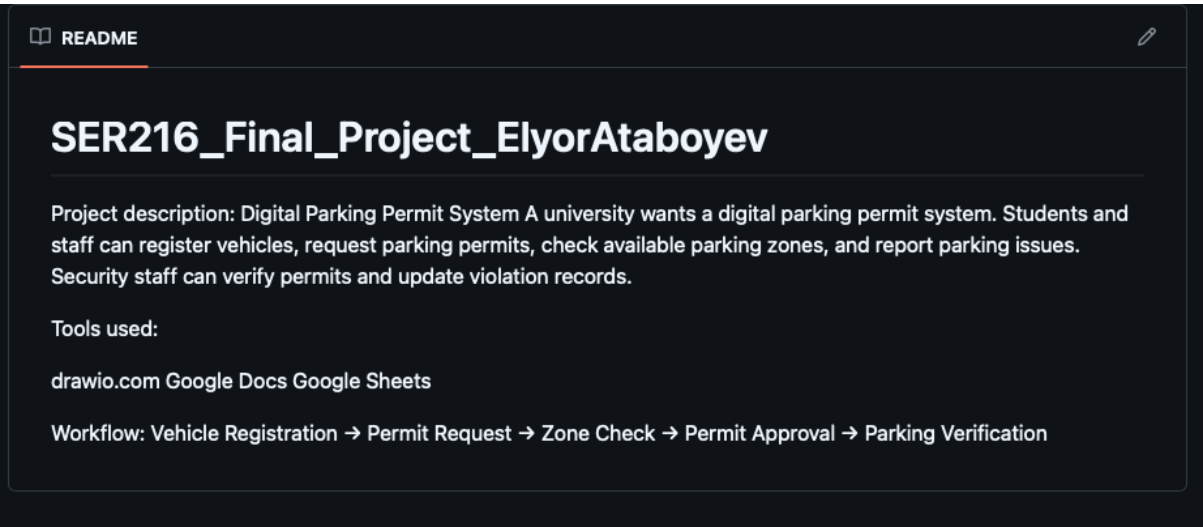
Item	Student Answer
Defect category	Functional / Logic Error
Severity	High (Prevents valid users from parking and generates false violations)
Priority	High (Must be fixed immediately before system launch/go-live)
Possible root cause	Timezone mismatch between the server database and the security scanner device, OR incorrect date comparison logic in the verification API (e.g., using expiration_date < current_date instead of <=, or comparing dates without accounting for time-of-day).
Regression test	TC-06: After applying the fix, scan a known valid, unexpired permit to confirm the system correctly displays "Valid" and does not trigger a false expiration flag.

Task 5: GitHub Evidence

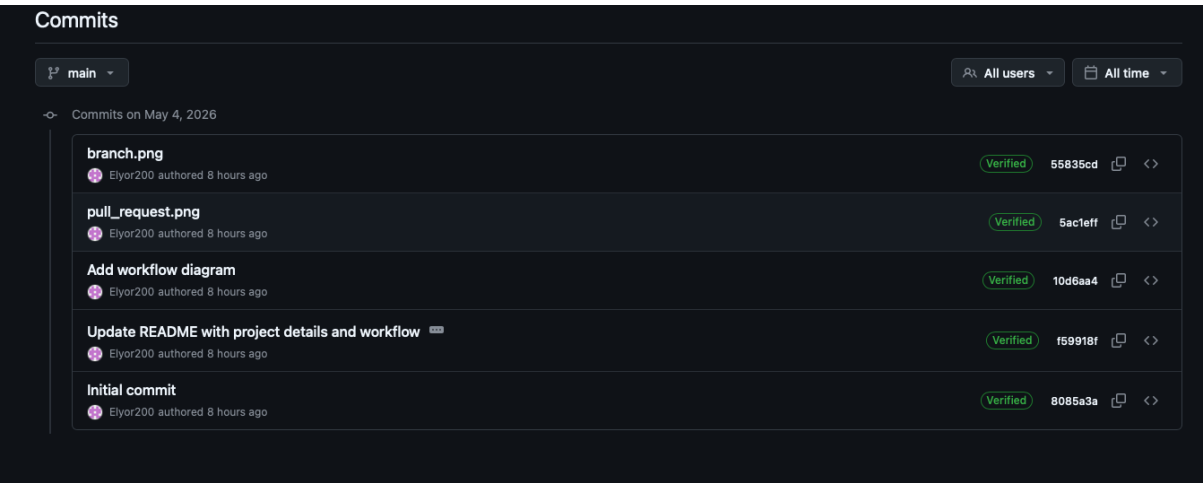
1. Repository Homepage



2. README.md File



3. Commit History (Showing at least 3 commits)



4. Feature Branch

Branches

New branch

OverviewYoursActiveStaleAll

Q Search branches...

Default

Branch	Updated	Check status	Behind	Ahead	Pull request
main	now				Default

Your branches

Branch	Updated	Check status	Behind	Ahead	Pull request
feature/change-request	3 minutes ago		1	1	
feature/test-plan	8 minutes ago		2	0	

Active branches

Branch	Updated	Check status	Behind	Ahead	Pull request
feature/change-request	3 minutes ago		1	1	
feature/test-plan	8 minutes ago		2	0	

5. Pull Request

Add workflow diagram #1

Merged Elyor200 merged 1 commit into feature/change-request from main 1 minute ago

Conversation 0Commits 1Checks 0Files changed 1

Elyor200 commented 1 minute ago

No description provided.

Add workflow diagram

Elyor200 merged commit 7f357c8 into feature/change-request 1 minute ago

Pull request successfully merged and closed

You're all set — the branch has been merged.

Add a comment

WritePreview

Add your comment here...

Markdown is supportedPaste, drop, or click to add files

Comment

6. Merged Pull Request

The screenshot shows a GitHub pull request titled "Add workflow diagram #1". The status is "Merged", indicating that Elyor200 has merged 1 commit into the `feature/change-request` branch from the `main` branch, 1 minute ago. The interface includes tabs for Conversation (0), Commits (1), Checks (0), and Files changed (1). A comment by Elyor200, posted 1 minute ago, states "No description provided." Below the comment, a commit titled "Add workflow diagram" is shown as "Verified" with commit hash 10d6aa4. A subsequent message confirms that Elyor200 merged commit 7f357c8 into feature/change-request 1 minute ago, with a "Revert" button available. A green banner announces "Pull request successfully merged and closed", noting that the branch has been merged. At the bottom, there is a section to "Add a comment" with a text area, a rich text editor toolbar, and a "Comment" button. A footer note mentions the repository's GitHub Community Guidelines.

7. Uploaded Final PDF in Repository

The screenshot displays a GitHub repository interface. On the left, a sidebar shows a search bar "Filter files..." and a file named "Final_report.pdf" with a green plus icon. The main content area features a "Search within code" bar and a dropdown menu showing "Final_report.pdf" with a copy icon. Below this, a message states "Binary file not shown." The "Comments" section shows 0 comments and a "Comment" input field. At the bottom, a "Subscribe" button is accompanied by the text "You're not receiving notifications from this thread."